Summary in Graph

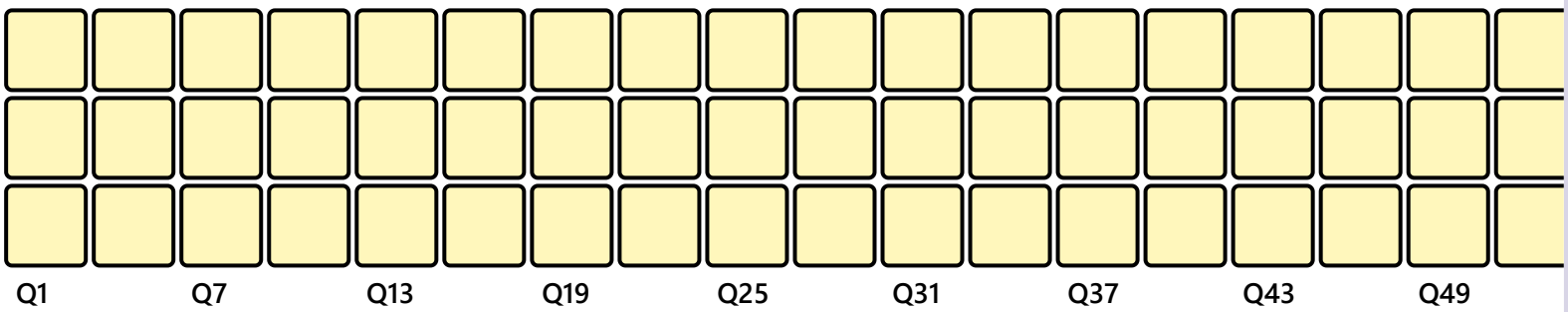
Exam Summary (GO Classes CS Test Series 2026 | MOCK GATE | Test 10)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (89) out of 893 users

EXAM RESPONSE EXAM STATS FEEDBACK

Aptitude

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Quantitative Aptitude

Lauren plays basketball with her friends. She makes 10 baskets. Each of these baskets is worth either 2 or 3 points. Lauren scores a total of 26 points. How many 3 point baskets did she make?

A. 5
B. 6
C. 7

D. 8

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

The digits A and B make the multiplication below correct. What is $A + B$?

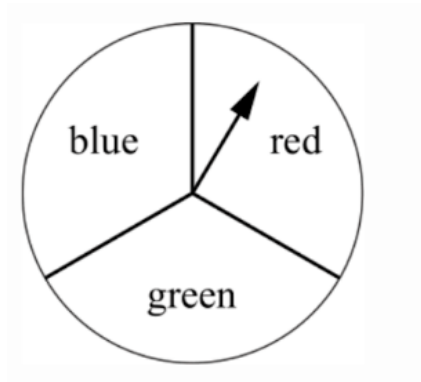
$$\begin{array}{r} 2\ A\ 7 \\ \times \quad\ \ 3 \\ \hline 7\ 1\ B \end{array}$$

A. 4
B. 6
C. 7
D. 9

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

A spinner is divided into 3 equal sections, as shown.



An arrow is attached to the centre of the spinner. The arrow is spun twice. What is the probability that the arrow lands on the same colour twice?

A. $\frac{1}{9}$

B. $\frac{2}{3}$

C. $\frac{1}{2}$

D. $\frac{1}{3}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

Kathy owns more cats than Alice and more dogs than Bruce. Alice owns more dogs than Kathy and fewer cats than Bruce. Which of the statements must be true?

A. Bruce owns the fewest cats
B. Bruce owns the most cats.

C. Kathy owns the most cats.
D. Alice owns the most dogs.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Choose the word from the questions given below that is most nearly opposite in meaning to the given word:
Frequency

A. periodicity
B. rarity
C. gradualness
D. persistency

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 02sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

In a magic garden, flowers can change during the night. If there is a group of at least two flowers of the same type directly beside each other, then that group of flowers transforms into a group of different flowers as follows:

- Groups of transform to .
- Groups of transform to .
- Groups of transform to .

A flower never changes more than once per night.
This is what the magic garden looks like one morning:

What will the magic garden look like after four nights have passed?

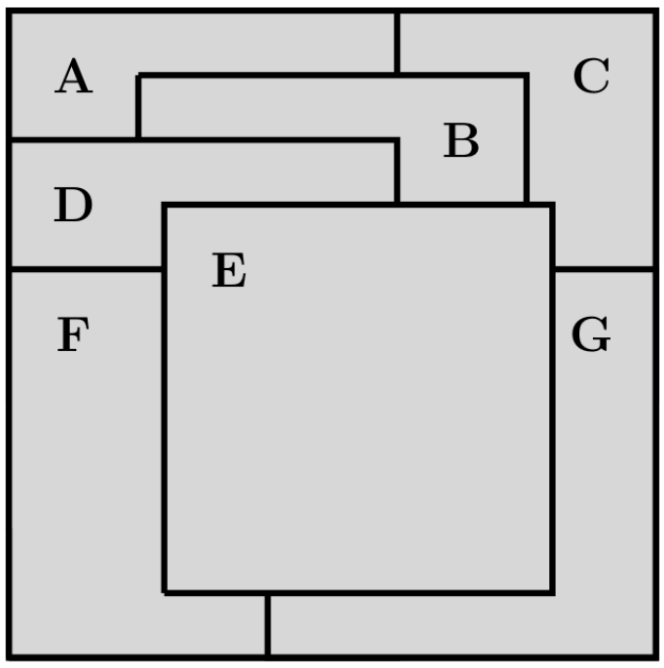
A.
B.
C.
D.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Spatial Aptitude

Seven identical square pieces of paper were labelled with the letters A through G. These papers were then placed on a table one at a time, in some order, ending with the paper labelled E. The diagram shows the view from above. In what order were the papers placed on the table?

Please note that E is placed on top and is completely visible, and all the other papers are of the same size and identical squares.

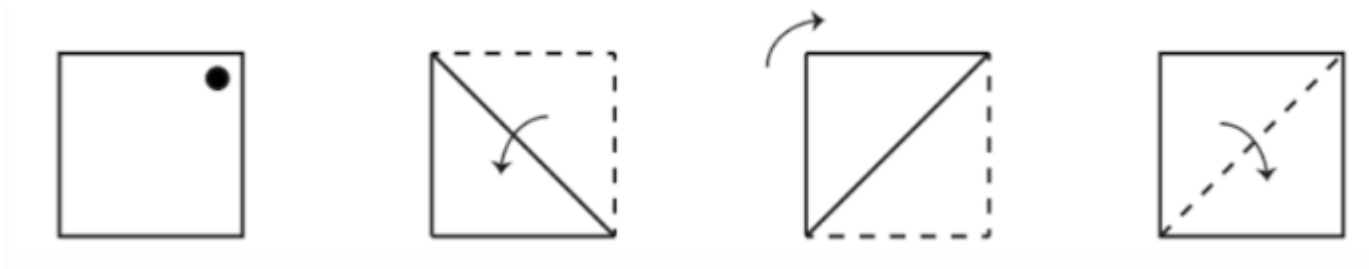


A. C, A, D, B, G, F, E
B. C, A, B, D, F, G, E
C. F, G, C, A, D, B, E
D. D, C, A, B, F, G, E

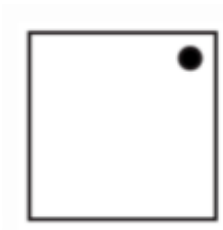
Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

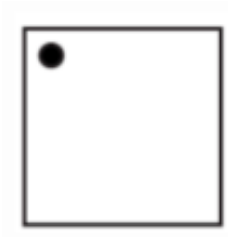
Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Spatial Aptitude

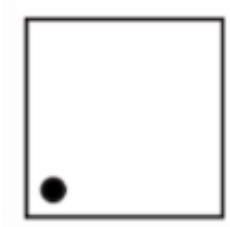
A square piece of paper has a dot in its top right corner and is lying on a table. The square is folded along its diagonal, then rotated 90° clockwise about its centre, and then finally unfolded, as shown.

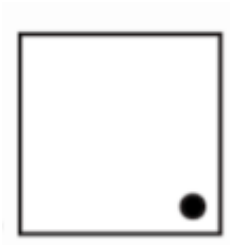


The resulting figure is:

A. 

B. 

C. 

D. 

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

The remainder when 111 is divided by 10 is 1. The remainder when 111 is divided by the positive integer n is 6. The number of possible values of n is _____.

A. 5
B. 8
C. 7
D. 6

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

One of the parts (A, B, C, D) in the sentence given below contains an ERROR. Which one of the following is **INCORRECT?**

I requested that he should be given the driving test today instead of tomorrow.

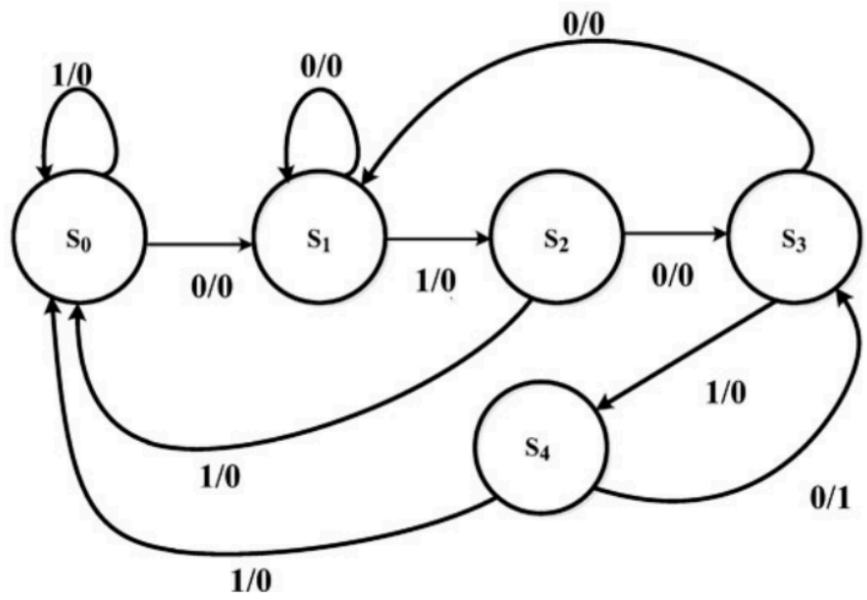
A. requested that
B. should be given
C. the driving test
D. instead of tomorrow

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

The state diagram of a sequence detector is shown below. State S_0 is the initial state of the sequence detector. If the output is 1, then

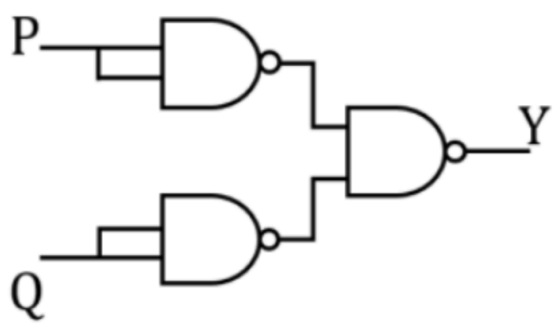


A. the sequence 01010 is detected.
B. the sequence 01011 is detected.
C. the sequence 01110 is detected.
D. the sequence 01001 is detected.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

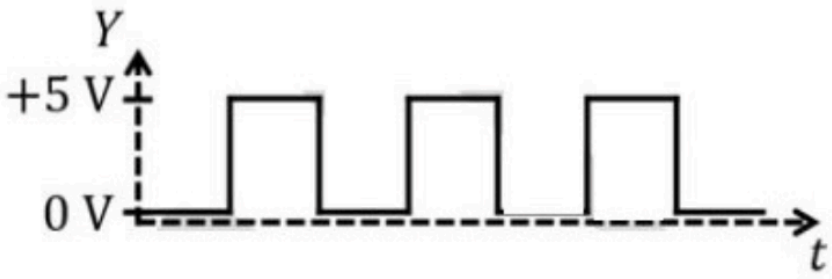
Consider the following circuit:

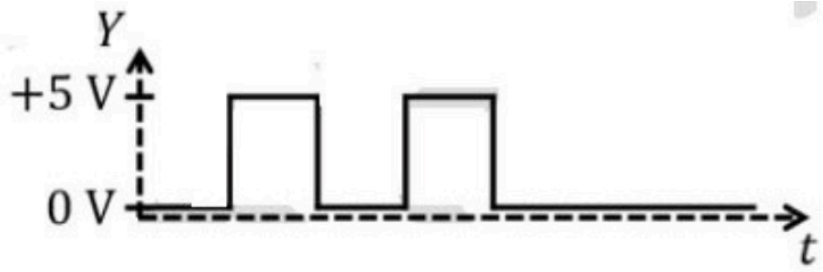


Suppose the input signal P is

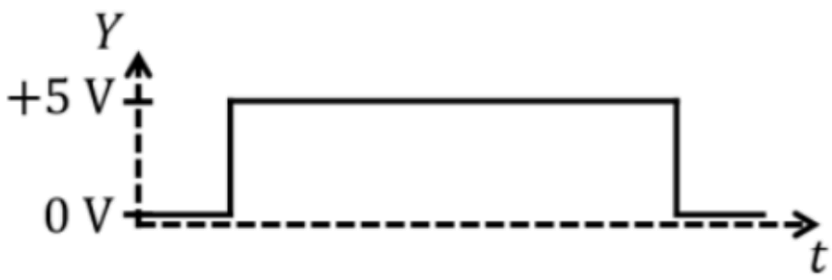
and the input signal Q is

Which one of the following output signals is correct?

A. 

B. 

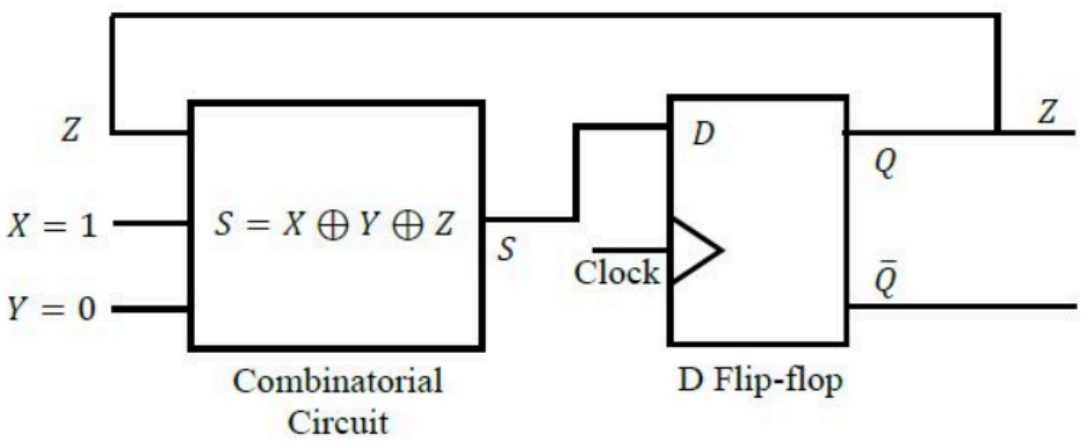
C.

D. 

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

In the circuit, the present value of Z is 1. Neglecting the delay in the combinatorial circuit, the values of S and Z, respectively, after the application of the clock will be



A. $S = 0, Z = 0$
B. $S = 0, Z = 1$
C. $S = 1, Z = 0$
D. $S = 1, Z = 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Consider the given statements below about TCP -

a. Host A is sending a large file to host B over a TCP connection. Assume host B has no data to send to host A. Host B will not send acknowledgments to host A because host B cannot piggyback the acknowledgments on data.

b. The size of the TCP *RcvWindow* never changes throughout the duration of the connection.

c. Suppose host A is sending a large file to host B over a TCP connection. The number of unacknowledged bytes that A sends cannot exceed the size of the advertised receiver buffer.

Which of the following statements is/are true?

A. Only a is true
B. a, b are true
C. All are true
D. Only c is true

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

The matrix

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

has

A. Eigenvalue $\lambda = 0$ with algebraic multiplicity 4 and geometric multiplicity 4 (i.e., 4 linearly independent eigenvectors).
B. Eigenvalue $\lambda = 0$ with algebraic multiplicity 4 and geometric multiplicity 2 (i.e., 2 linearly independent eigenvectors).
C. Eigenvalue $\lambda = 0$ with algebraic multiplicity 2 and geometric multiplicity 2 (i.e., 2 linearly independent eigenvectors).
D. Eigenvalue $\lambda = 0$ with algebraic multiplicity 2 and geometric multiplicity 1.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be the output of the following program?

It is given that sizeof(int) $=$ 4 and sizeof(int *) $=$ 8.

```
#include<stdio.h>
main(){
    void** p = 1000;
    int* q = ((int*)(p + 1)) - 1;
    printf("%u", q);
}
```

A. 1000
B. 1008
C. 1004
D. Error, since $p + 1$ is not allowed as p is declared as void.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Consider set of four $/24$ IP network addresses as follow as-

$$212.56.146.0/24$$
$$212.56.147.0/24$$
$$212.56.148.0/24$$
$$212.56.149.0/24$$

Suppose we want to do address aggregation and we are not allowed to include any extra address except address in above four networks.
Which of the following prefixes in CIDR notation can be collectively used to correctly aggregate all of the subnets in the routing table?

A. $212.56.146.0/23$
B. $222.56.144.0/22$
C. $212.56.148.0/23$
D. $222.56.147.0/23$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Suppose that three processes exist in the ready state of the system, as described table below.
Suppose that the system uses preemptive, round-robin scheduling and that T_{11} is running when the quantum expires.

Process	Threads within the process
P_1	T_{11}, T_{12}
P_2	T_{21}, T_{22}
P_3	T_{31}

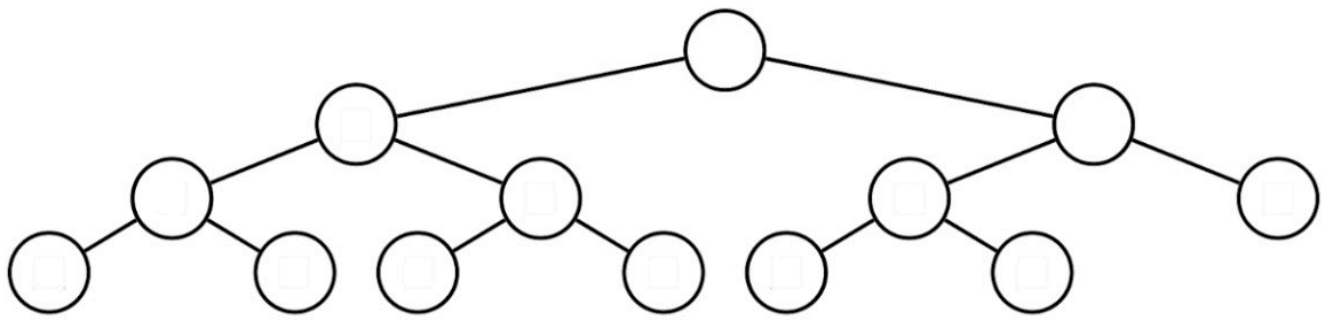
If the threads are implemented entirely at the user level (with no support from the operating system), which threads might possibly execute at the beginning of the next quantum?

A. T_{12}
B. T_{21}
C. T_{22}
D. T_{31}

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

If the largest value in a max-heap is in the root node and the second-largest value is in the root's right child, what will be the number of positions where the fourth-largest value can be put? Assume all values are distinct.



Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

The power set $\mathcal{P}((\mathrm{A} \times \mathrm{B}) \cup (\mathrm{B} \times \mathrm{A}))$ has the same number of elements as the power set $\mathcal{P}((\mathrm{A} \times \mathrm{B}) \cup (\mathrm{A} \times \mathrm{B}))$ if and only if

A. $\mathrm{A} = \mathrm{B}$
B. $\mathrm{A} = \emptyset$ or $\mathrm{B} = \emptyset$
C. $\mathrm{B} = \emptyset$ or $\mathrm{A} = \mathrm{B}$
D. $\mathrm{A} = \emptyset$ or $\mathrm{B} = \emptyset$ or $\mathrm{A} = \mathrm{B}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

In a group G, every element other than the identity element has order 2 then G is?

A. Abelian group
B. Cyclic group
C. Non-abelian group
D. Non-cyclic group

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Consider the premises below which are assumed to be true:

$$p \rightarrow (q \wedge r)$$
$$\sim p \rightarrow \sim r$$
$$\sim p \wedge (r \rightarrow \sim p)$$

Which of the following conclusions would make the argument valid?

i. $p \rightarrow q$
ii. $\sim p \vee \sim r$
iii. $q \rightarrow (p \rightarrow r)$
iv. $\sim p \wedge \sim q \wedge r$

A. Only (i) and (iii).
B. Only (ii) and (iv).

C. Only (i), (ii) and (iii).
D. Only (i), (ii) and (iv).

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

The minimal finite automata that accepts all strings of a's and b's, where the number of a's is at least 'n' contains

A. n states
B. $(n+1)$ states
C. $(n+2)$ states
D. $(n+3)$ states

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

$$\mathrm{S} \rightarrow a\mathrm{S}b \mid b\mathrm{S}a \mid \mathrm{S}\ \mathrm{S} \mid \epsilon$$

Which of the following best characterizes the language generated by the grammar above?

A. All strings of the form $a^i b^j a^k$, where $i+j=k$
B. All palindromes over a and b
C. All strings with equal numbers of a 's and b 's
D. All strings of the form ww^{R}, where $w \epsilon \{a, b\}^*$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

A machine has a 32-bit architecture with 1-word long instructions. It has 24 registers and supports an instruction set of size 40. Each instruction has five distinct fields, namely opcode, two source register identifiers, one destination register identifier, and an immediate value. Assuming that the immediate operand is an unsigned integer, its maximum value is _____.

Your Answer: | Correct Answer: 2047 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Consider a disk with the following characteristics :

- Average seek time $= 8$ msec
- Average rotational delay $= 3$ msec
- Maximum rotational delay $= 6$ msecs
- Spindle speed $= 10,000$ rpm
- Sectors per track $= 170$ sectors
- Sector size $= 512$ bytes

What is the average time to read one sector?

A. 10
B. 11.03
C. 12.35
D. 14.22

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Numerical Type | Award: 1 | Penalty: 0 | Compiler Design

Consider the following three-address code (line numbers added):

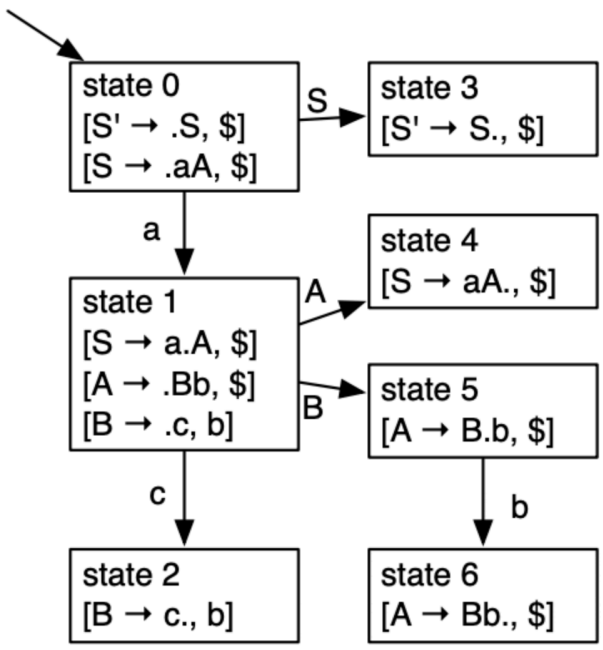
```
1   L1:   i := a
2         goto L5
3   L2:   s := s + 1
4         goto L5
5   L3:   if j <= i then goto L1
6   L4:   i := i + 1
7         goto L1
8   L5:   n := n - 1
9         if n > 0 then goto L3
10
```

For the above piece of code, the number of unreachable basic blocks in the Control Flow Graph (CFG) are?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

Consider the following $\mathrm{LR}(1)$ parsing DFA.



Suppose that the $\mathrm{LR}(1)$ stack contains ac at some stage of parsing.

What action is performed when the input symbol is b?

A. Shift
B. Reduce by $\mathrm{B} \rightarrow \mathrm{c}$
C. Reduce by $\mathrm{A} \rightarrow \mathrm{Bb}$
D. Accept

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider classic dinning philosopher problem with arbitrary number of D diners. Furthermore, each diner $d = 1 \ldots \mathrm{D}$ may require a different number of S_d spoons to eat. For example, it is possible that one of the diners is an octopus, who for some reason refuses to begin eating before acquiring $\mathrm{S}_{\text{octopus}} = 8$ spoons. What is the smallest number of spoons needed to ensure that deadlock can not occur?

A. $\sum_d \mathrm{S}_d - \mathrm{D} + 1$
B. $\sum_d \mathrm{S}_d + \mathrm{D}$
C. $\sum_d \mathrm{S}_d - 1$
D. $\sum_d \mathrm{S}_d - \mathrm{D}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

For all asymptotically positive functions $f(n)$, $f(n) + o(f(n))$ (where o denotes small-o notation) is equal to one of the following?

A. $\Theta(f(n))$
B. $\Theta\left(f(n)^2\right)$
C. $o(f(n))$ (where o denotes small-o notation)
D. $\Theta(f(n) \log n)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Numerical Type | Award: 1 | Penalty: 0 | Algorithms

Let $G(V, E)$ be an undirected graph with 9 vertices. Each edge has a weight of either 1 or 2. Let E' be the set of edges with weight 1, and assume that $G' = (V, E')$ is the subgraph containing only the edges with weight 1. This subgraph has two connected components. Based on this information, what is the weight of the minimum spanning tree (MST) in G?

Your Answer: | Correct Answer: 9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Recall the 3-valued logic for evaluating NULL in SQL.

Suppose A $=$ NULL, B $=$ 5, C $=$ 10.

What will be the value of

$$((A < 7) \quad OR \quad (C > 2)) \quad AND \quad (B > 10)$$

A. True
B. False
C. Null
D. Unknown

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Which of the following statements is NOT true?

A. If $A \subset B$, then $\mathbf{P}(A) \leq \mathbf{P}(B)$.
B. If $\mathbf{P}(B) > 0$, then $\mathbf{P}(A \mid B) \geq \mathbf{P}(A)$.
C. $\mathbf{P}(A \cap B) \geq \mathbf{P}(A) + \mathbf{P}(B) - 1$.
D. $\mathbf{P}\left(A \cap B^c\right) = \mathbf{P}(A \cup B) - \mathbf{P}(B)$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Assume a shortest Remaining Time First (SRTF) scheduler for the original workload with jobs A, B, and C. Assume a new job D requiring 5 seconds of CPU time arrives at some time T.

Job Name	Arrival Time (seconds)	CPU Burst Time (seconds)
A	0	8
B	2	4
C	5	7

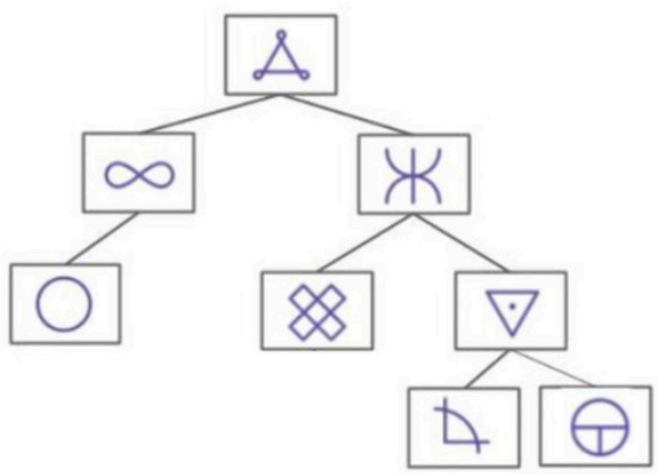
For which arrival times of T would D preempt the job running at that time?

A. 3 seconds
B. 5 seconds
C. 8 seconds
D. 13 seconds

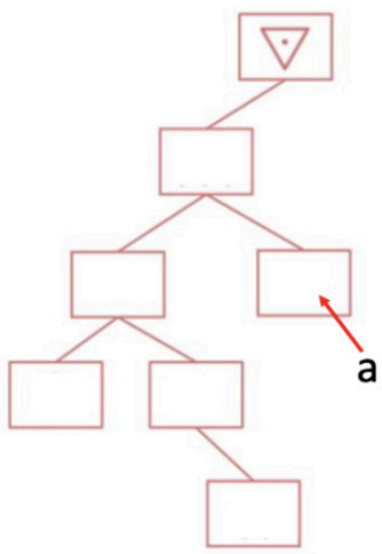
Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Consider the following Binary search tree, some objects and the empty structured binary search tree shown. Assume no two objects are equal.



Binary Search Tree (Filled)



Binary Search Tree (Empty Structure)

What could be the possible object(s) in place of **"a"** as indicated in the above figure?

A.

B.

C.

D.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider a relation $\mathrm{R(T, U, V, W, X, Y)}$ with functional dependencies\\

$$\{\mathrm{T} \rightarrow \mathrm{UV}, \mathrm{TV} \rightarrow \mathrm{WX}, \mathrm{UX} \rightarrow \mathrm{Y}\}.$$

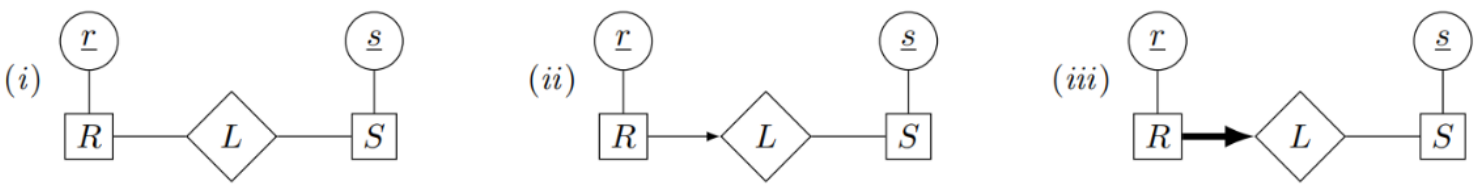
Which of the following is/are lossless join decompositions of R into BCNF?

A. TVW and UXY
B. T and $UVWXY$
C. $TUVWX$ and UXY
D. $TUVWX$ and $TUXY$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following ER diagrams labelled $i, ii,$ and iii.



Which ER diagram do the following SQL statements correspond to?

```
CREATE TABLE S(s INT, PRIMARY KEY (s));
CREATE TABLE R(r INT, s INT NOT NULL REFERENCES S, PRIMARY KEY (r));
```

Note that there is exactly one correct answer. Write the answer as 1 or 2 or 3 for the correct ER diagram.

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Two transactions are "not-interleaved" in a schedule if every action of one transaction precedes every action of the other.
We say that a transaction T precedes a transaction U in a schedule S if they are not interleaved, and all actions of T precedes all actions of U in S.

Note that if T and U are the Only transactions in S, then saying T precedes U is the same as saying that S is the serial schedule (T, U).

However, if S involves transactions other than T and U, then S might not be a serial schedule, in fact, because of the effect of other transactions, S might not even be conflict-serializable.

Consider a conflict-serializable schedule S in which transactions $\mathrm{T1}$ and $T2$ are not interleaved in S, and $T1$ precedes $T2$ in S. The Schedule S may include more than 2 transactions.

Which of the following must be true for S?

A. In any serial schedule conflict equivalent to $\mathrm{S}, \mathrm{T1}$ precedes $T2$.
B. In any serial schedule conflict equivalent to $\mathrm{S}, \mathrm{T2}$ precedes $T1$.
C. There exists at least one serial schedule conflict equivalent to S in which $T1$ precedes $T2$.
D. None of the above is necessarily true.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

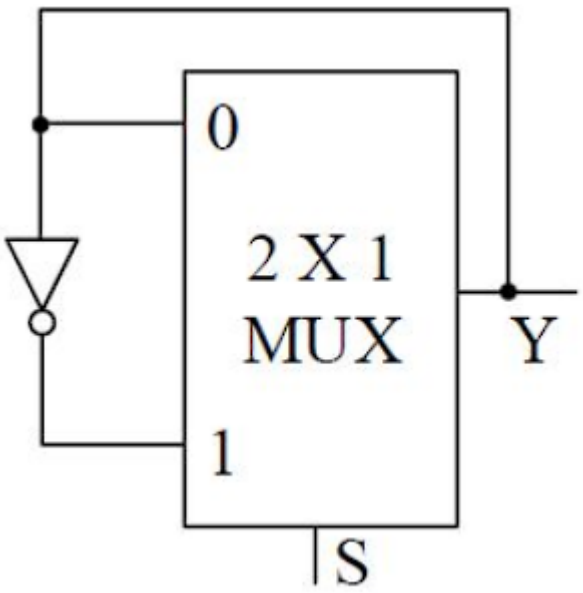
Q #29 | Numerical Type | Award: 2 | Penalty: 0 | Databases

A $\mathrm{B\text{-}Tree}$ of order m is an m-way search tree in which every node, except root node, has at least $m/2$ children. The height of a one-level tree is defined to be 0, the height of a tree with two levels is defined to be 1, and so on. What is the minimum number of distinct keys (or records) stored in a $\mathrm{B\ Tree}$ of height 5 and order 5?

Your Answer: | Correct Answer: 485 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The propagation delay of the 2×1 MUX shown in the circuit is $10\ \mathrm{ns}$. Consider the propagation delay of the inverter as $0\ \mathrm{ns}$.



If S is set to 1 then the output Y is _____.

A. a square wave of frequency $100\mathrm{MHz}$
B. a square wave of frequency $50\mathrm{MHz}$
C. constant at 0
D. constant at 1

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Traffic lights display three basic colors: red (R), yellow (Y), and green (G), so we can use them to encode base 3! We decide to use the encoding $0 \leftrightarrow R, 1 \leftrightarrow Y, 2 \leftrightarrow G$. For example, $5 = 1 \times 3^1 + 2 \times 3^0$ would be encoded as **YG.** Assume each traffic light can only display one color at a time.

If we have 9 bits of binary data that we want to store, minimum how many traffic lights would it take to store that same data?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

For any language L, We define middle $(\mathrm{L}) = \{x : \exists y, z \in \Sigma^*, (yxz \in \mathrm{L})\}$.

Then middle $(\{w \in \{a, b\}^*, \#(a) = \#(b)\})$ is _____.

A. Regular.
B. DCFL but Not regular.
C. CFL but Not DCFL.
D. Not CFL.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Consider the definition of the Deterministic Turing machine in which we have a designated accept state q and a reject state r, where q and r are different states. The computation halts (halts immediately) if and only if the Turing machine enters one of these two states.
Suppose M is a Turing machine that recognizes the Language L.

Let M' be a Turing machine that is identical to M except the accept and reject states are swapped. Then which of the following is/are viable possibilities (i.e. is possible to be true for some M) for $\mathrm{L}(\mathrm{M}), \mathrm{L}\left(\mathrm{M}'\right)$?

A. $\mathrm{L}(\mathrm{M}) = \mathrm{L}\left(\mathrm{M}'\right)$
B. $\mathrm{L}(\mathrm{M})! = \mathrm{L}\left(\mathrm{M}'\right)$
C. $\mathrm{L}(\mathrm{M})$ is complement of $\mathrm{L}\left(\mathrm{M}'\right)$
D. $\mathrm{L}(\mathrm{M}) \cap \mathrm{L}\left(\mathrm{M}'\right) = \Phi$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

For any languages L, L', which of the following is/are false statements?(Note that L' is not the notation for complement of L)

A. If L and L' are not regular, then $L \cup L'$ is not regular.
B. If L is regular and $L \cup L'$ is regular, then L' is regular.
C. If L is regular and $L \cup L'$ is not regular, then L' is not regular.
D. If L is not regular and $L \cup L'$ is regular, then L' is regular.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

In a certain implementation of strings, characters are 8 bits long and words are 36 bits long. Characters are packed into double words so that no space is lost. How many words of memory does a k-character string require? (If the string does not occupy an integral number of double words, enough binary zeros are appended to reach the next single word boundary.)

A. $\lfloor 2k/9 \rfloor$
B. $\lfloor k/9 \rfloor$
C. $2\lceil k/9 \rceil$
D. $\lceil 2k/9 \rceil$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Determine the average branch penalty for a branch-target buffer, assuming the penalty cycles for individual mispredictions is as following :

- When instructions are in buffer, and if prediction is correct then no penalty cycles.
- When instructions are in buffer, and if prediction is incorrect then 2 penalty cycles.
- When instructions are Not in buffer, and if branch is taken then 2 penalty cycles.
- When instructions are Not in buffer, and if branch is not taken then 0 penalty cycles.

Among the branches which are not in buffer only 60% are taken. The prediction accuracy is 90% and hit rate in the buffer is 90%.

What is the average branch penalty (upto two digits after decimal point)?

Your Answer: | Correct Answer: 0.30 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

An inversion in an array A of numbers is any ordered pair (i, j) such that $iA[j]. $Consideranarray$A $oflength$100. $Whatisthemaximumnumberofinversionsthatcanberemovedbyexchanging$A[10]$and$ A[30]?$

Your Answer: | Correct Answer: 39 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

In a game where two players take turns picking cards from either end of a sequence $s_1, s_2, \ldots, s_n$, each card s_i has a value v_i. Both players aim to maximize their total value and play optimally.

Define $V(i, j)$ as the maximum score the first player can achieve on a sub-sequence from index i to j, aiming to maximize their score while minimizing the opponent's.

A purely greedy approach (always taking the highest visible value) may fail in sequences like $4, 9, 1, 3$, where taking the 4 exposes the 9 to the opponent.

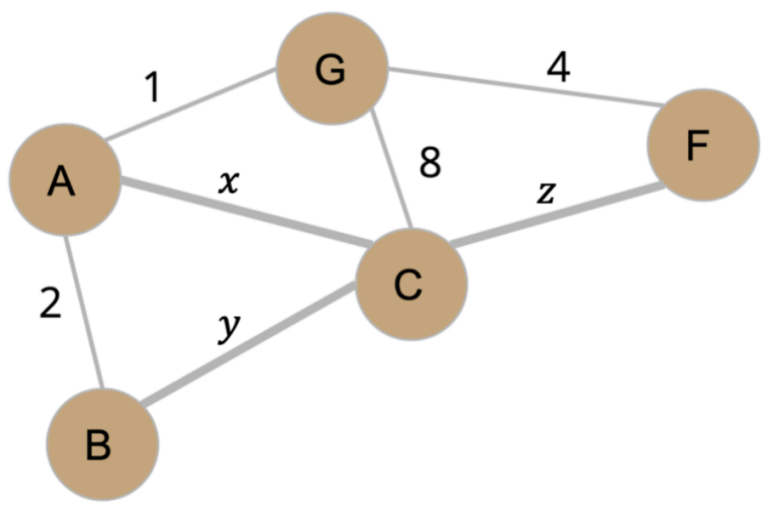
Which of the following equations correctly describes $V(i, j)$?

A. $V(i, j) = \max\left(v_i + V(i+1, j), v_j + V(i, j-1)\right)$
B. $V(i, j) = \max\left(v_i - V(i+1, j), v_j - V(i, j-1)\right)$
C. $V(i, j) = \max\left(V(i+1, j) - v_i, V(i, j-1) - v_j\right)$
D. $V(i, j) = \min(V(i+1, j), V(i, j-1))$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

$\mathrm{G(V,E)}$ is the graph given below where costs of few edges incident on node C are unknown.



We run Prim's algorithm on G starting with vertex A.

- Prim's algorithm grows a tree at every step and gives MST at the last step. Initially we have only A in tree and based on edge weights Prim's algorithm will include other vertices one at a time.
- Prim's algorithm uses a priority queue to maintain the priority of every vertex, and it picks the minimum priority vertex at every step.

What should be the relationship between x, y and z such that we include C in MST as last vertex and the priority of C changes the maximum number of times?

A. $y > 4$
B. $x > \mathrm{GC} > y > z$
C. $y > 8$
D. $x > y > z > \mathrm{GC}$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider the following singly linked list where the head points to the first node of the linked list.

$$2 \rightarrow 4 \rightarrow 6 \rightarrow 8 \rightarrow 10 \rightarrow \text{NULL}$$

If we execute the following code on the above-linked list then what will be the final linked list?

```
Node* recursive_swap(Node* head) {
    if (head == NULL || head->next == NULL)
        return head;

    Node* first = head;
    Node* second = head->next;

    first->next = recursive_swap(second->next);
    second->next = first;

    return second;
}
```

A. $2 \rightarrow 4 \rightarrow 6 \rightarrow 8 \rightarrow 10 \rightarrow$ NULL
B. $4 \rightarrow 2 \rightarrow 8 \rightarrow 6 \rightarrow 10 \rightarrow$ NULL
C. $2 \rightarrow 6 \rightarrow 4 \rightarrow 8 \rightarrow 10 \rightarrow$ NULL
D. $4 \rightarrow 2 \rightarrow 6 \rightarrow 8 \rightarrow 10 \rightarrow$ NULL

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

Consider following C code -

```
int *pi1 = (int *) malloc(2 * sizeof(int));
int *pi2 = pi1 + 1;
int i = 1;
free(pi1);
*(pi2 - 1) = i;
pi2 = NULL;
```

What memory management error(s) result from each code snippet?

A. Dangling pointer but No memory leak
B. Memory leak but No Dangling pointer
C. Both Dangling pointer and Memory leak
D. None

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

In C programming, constant integers are considered to be signed integers by default. One way to represent them as an unsigned constant is by appending U as a suffix. For example, -1 is signed whereas $-1U$ is unsigned.

Which of the following condition(s) is/are TRUE?

A. $-1 > -2$
B. $-1U > -2$
C. $-1 > 0U$
D. $-1 > 0$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the given grammar below:

$$E \rightarrow id \mid (E) \mid EE$$

How many times do we need to backtrack to parse the string "(id)((id))" using recursive descent backtrack parser?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

Suppose we take the disjoint union of two copies of K_n; let's say that one of them has vertex set $\{v_1, v_2, \ldots, v_n\}$ as usual, but the other one has vertex set $\{w_1, w_2, \ldots, w_n\}$ instead. Then, we take the complement. What is the chromatic number of this new graph that we get, if $\mathrm{n} = 10$?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 Numerical Type | Award: 2 | Penalty: 0 | Combinatory

A message is a sequence of two types of signals: of type A which last for 1 second and of type B which last for 2 seconds. E.g., the message ABAAB lasts 7 seconds. How many different messages last 10 seconds?

Your Answer: | Correct Answer: 89 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

A processor uses 30 bit physical address and 32-bit virtual address.

We subdivide the virtual address into 4 segments as follows:

10-bit	8-bit	6-bit	8 bit

We use a 3-level page table, such that the first 10-bit are for the first level and so on. Each page table entry is of size 4 bytes.

The number of bits required for addressing the next level page table(or page frame) in the page table entry of the first, second and third level page tables are respectively

A. $22, 23, 24$
B. $22, 22, 22$
C. $20, 22, 22$

D. $20, 22, 24$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider a Unix-like file system that maintains a unique index node (inode) for each file in the system. Each index node includes 6 direct pointers, a single indirect pointer, and a double indirect pointer. The file system block size is B bytes, and a block pointer occupies P bytes. How many disk operations are required if a process reads data from the Nth block of a file? Assume the file is already open, the buffer cache is empty, and each disk operation reads a single file block.

A. 1 disk operation if $N \leq 6$
B. 2 disk operations if $6 < N \leq 6 + B/P$
C. 3 disk operations if $N > 6 + B/P$
D. 2 disk operations for all N

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Three processes coordinate their execution using shared semaphores to print the word **HELLO**. Execution may switch between any process at any time.

Select the correct semaphore initializations and placements of *wait(S)* and *signal (S)* to ensure the word **HELLO** is printed exactly once.

A) Semaphores: a = 1, b = 0, c = 0

Process 1

```
Loop1:
    wait(a)
    print("H")
    print("E")
    signal(b)
    signal(b)
    goto Loop1
```

Process 2

```
Loop2:
    wait(b)
    print("L")
    signal(c)
    goto Loop2
```

Process 3

```
Loop3:
    wait(c)
    wait(c)
    print("O")
    goto Loop3
```

B) Semaphores: a = 1, b = 1, c = 0

Process 1

```
Loop1:
    wait(a)
    print("H")
    print("E")
    signal(b)
    goto Loop1
```

Process 2

```
Loop2:
    wait(b)
    print("L")
    signal(c)
    goto Loop2
```

Process 3

```
Loop3:
    wait(c)
    print("O")
    goto Loop3
```

C) Semaphores: a = 1, b = 0, c = 1

C.

Process 1

```
Loop1:
    wait(a)
    print("H")
    print("E")
    signal(b)
    goto Loop1
```

Process 2

```
Loop2:
    wait(b)
    print("L")
    signal(c)
    goto Loop2
```

Process 3

```
Loop3:
    wait(c)
    print("O")
    goto Loop3
```

D) Semaphores: a = 0, b = 1, c = 0

D.

Process 1

```
Loop1:
    wait(a)
    print("H")
    print("E")
    signal(b)
    goto Loop1
```

Process 2

```
Loop2:
    wait(b)
    print("L")
    signal(c)
    goto Loop2
```

Process 3

```
Loop3:
    wait(c)
    print("O")
    goto Loop3
```

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Host A sends a file consisting of 9 MSS-sized segments to a host B using TCP. Assume that the 4th segment in the transmission is lost and congestion is detected by 3 duplicate ACKs. 3 duplicate ACKs indicate that 3 segments has been reached but one of the early segment is missing.

Consider two versions of TCP:

- Version 1 : On receiving 3 duplicate ACKs, it goes to slow start with following parameters- ssthresh $= \mathrm{cwnd}/2$, cwnd $= 1$.
- Version 2 : On receiving 3 duplicate ACKs, it goes to congestion avoidance with following parameters – $\mathrm{ssthresh} = \mathrm{cwnd}/2,\ \mathrm{cwnd} = \mathrm{cwnd} + 1.$

Which of the following is/are true about the immediately next packets TCP will send after detection of congestion?

A. In Version 1, packet sent is $= 4$
B. In Version 1, packets sent are $= 4, 8$
C. In Version 2, packets sent are $= 4, 5, 6, 7, 8$
D. In Version 2, packets sent are $= 4, 8$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

The information bit sequence $\{111010101\}$ is to be transmitted by encoding with Cyclic Redundancy Check 4 (CRC-4) code, for which the generator polynomial is $C(x) = x^4 + x + 1$. The encoded sequence of bits is _____.

A. $\{1110101011100\}$
B. $\{1110101011101\}$
C. $\{1110101011110\}$
D. $\{1110101010100\}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 Numerical Type | Award: 2 | Penalty: 0 | Calculus

Evaluate the limit

$$\lim_{x \to \infty} \frac{\sqrt{4x^6 - x}}{x^3 + 8}$$

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the set of vectors $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} h+1 \\ h \\ h-2 \end{pmatrix}$$

and h is a parameter. Which statement is true?

A. $\mathcal{B}$ is a linearly independent set of vectors for all h
B. $\mathcal{B}$ is a linearly independent set of vectors exactly when $h = 0$
C. $\mathcal{B}$ is a linearly independent set of vectors exactly when $h \neq 5$
D. $\mathcal{B}$ is a linearly independent set of vectors exactly when $h \neq -1$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 Numerical Type | Award: 2 | Penalty: 0 | Programming in C

What would be displayed if the following code fragment was executed?

```
int n = 4;
int s = -n;
while (n > 1) {
    n = n - s;
    s = s + n;
    s++;
    n--;
}
printf("%d", s);
```

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 Multiple Select Type | Award: 2 | Penalty: 0 | Probability

We toss two fair coins simultaneously and independently. If the outcomes of the two coin tosses are the same, we win; otherwise, we lose. Let A be the event that the first coin comes up heads, B be the event that the second coin comes up heads, and C be the event that we win. Which of the following statements is/are true?

A. Events A and B are not independent.
B. Events A and C are independent.
C. Events A and B are conditionally independent given C.
D. The probability of winning is $3/4$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

An array **arr** of integers is always stored with an integer-aligned address (i.e., the base address is divisible by the size of an integer), whether in main memory or in the cache. Currently:

- The array resides in main memory.
- The cache is empty.

The cache is a single-level, direct-mapped cache with a block size of 256 bytes. Assume the size of an integer is 4 bytes.
If sequential accesses are performed to **arr**[0], **arr**[1], and **arr**[2], what is the probability of getting cache hits for **arr**[1] and **arr**[2] after the initial cache miss for **arr**[0]?

A. $63/64$
B. $3/4$
C. $31/32$
D. $1/2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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